* Revision:

① Population vs Sample:

② mean — median — mode

mean → Avg → an out

median → middle when in sorted → ∝

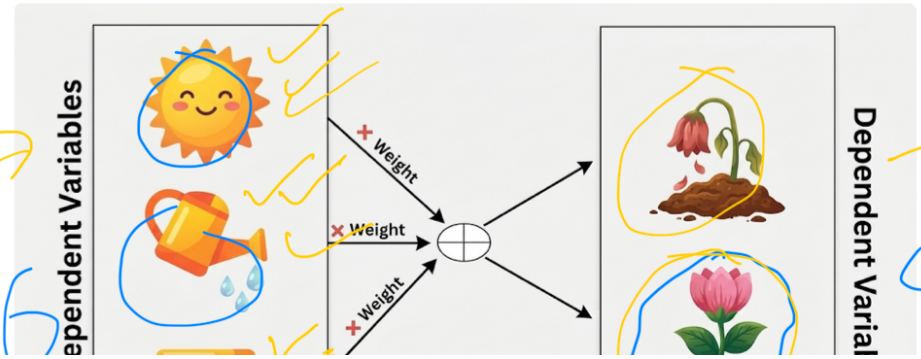
mode → for most the data

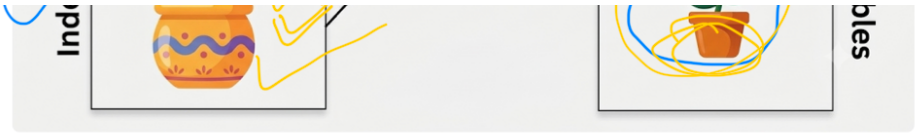
③ Range → 1 to 10

variance (σ^2) → $\sigma^2 = \frac{\Sigma (x-\mu)^2}{n}$

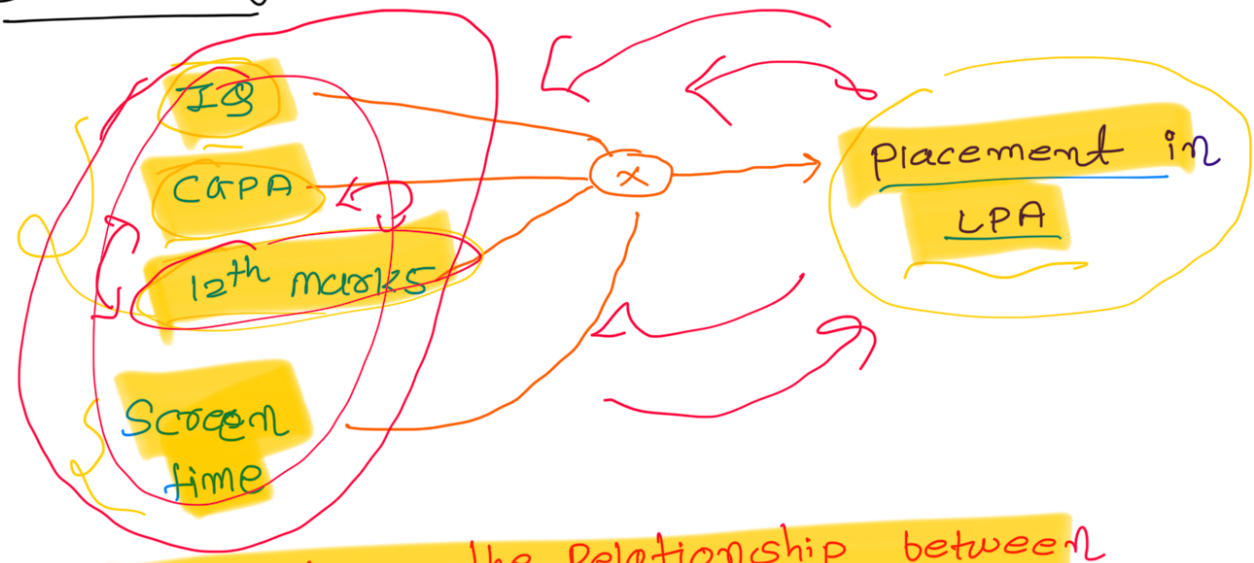
std (σ) → $\sqrt{\text{variance}}$

* Today's class:

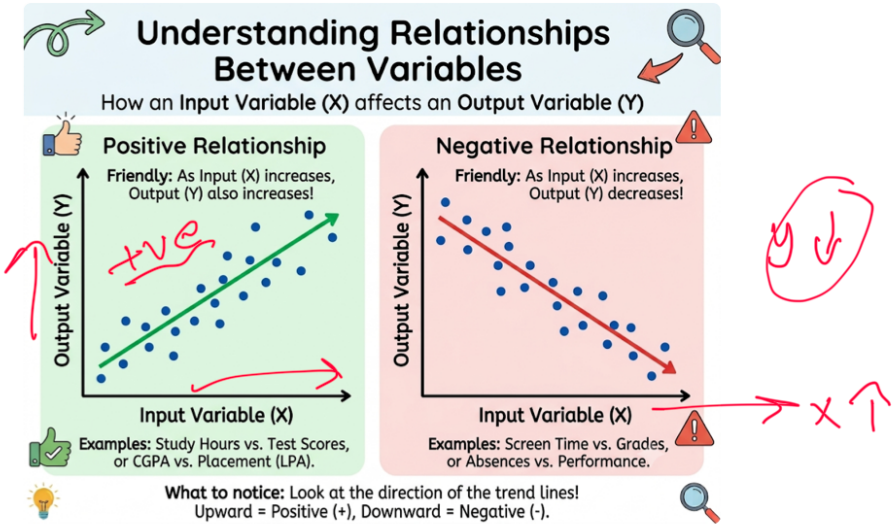




→ Same way :



→ If we know the Relationship between Input & output variables then It will help us Identify the Important features & eventually we can build better Model.

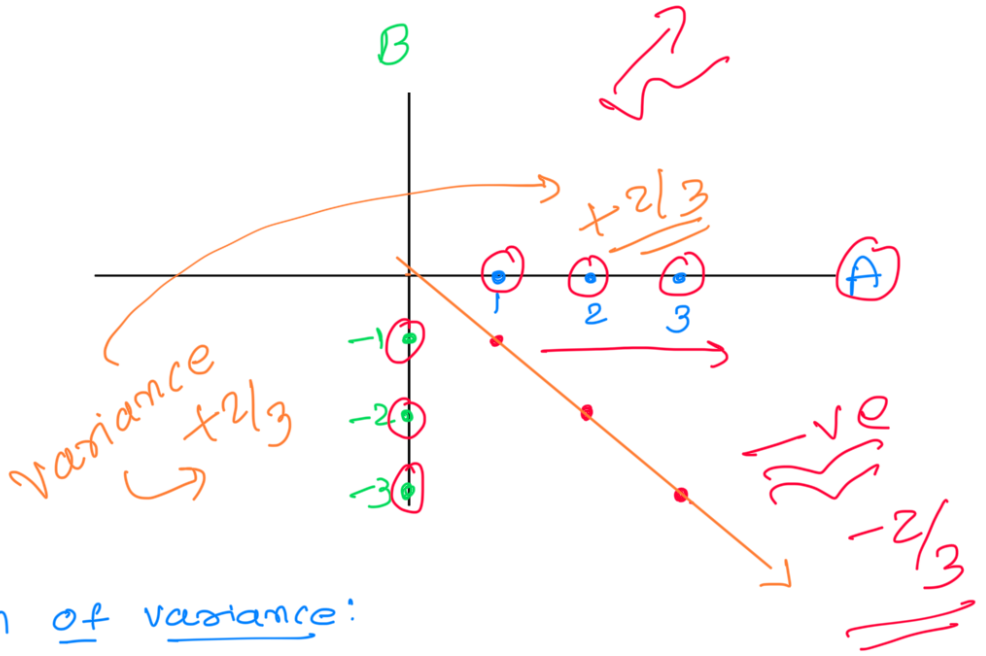


Columns

	X	y	A	B
R_1 →	-10	-20	-3	3
R_2 →	0	0	-2	2
				1

$R_3 \rightarrow$ 10 | 20 | −1 | 1

	R_1	R_2	R_3	mean	variance	std
X:	−10	0	10	0	200/3 = 66.6	
y:	−20	0	20	0	800/3 = 266.6	
A:	−3	−2	−1	−2	+2/3	
B:	3	2	1	2	+ 2/3	



* limitation of variance:

→ No Directional Info using Variance.

→ variance calculte the spread of one variable & can't give us the Relationship info between two variables.

→ Var is unscaled scale.

① Covariance.

$$cov(x,y) = \Sigma (x - \mu_x)(y - \mu_y) \qquad \Sigma (x - \mu_x)^2$$

η

η

	R_1	R_2	R_3	mean	variance	covariance
A:	-3	-2	-1	-2	$+^2/_3$	
B:	3	2	1	2	$+^2/_3$	

$$\frac{(-3-(-2))(3-2)+(-2-(-2))(2-2)+(-1-(-2))(1-2)}{3}$$

$$= \frac{(-3+2)(1)+(-2+2)(2-2)+(-1+2)(-1)}{3}$$

$$= \frac{-1+0-1}{3}$$

$$= -2/3$$

covariance

$$\frac{\Sigma (x-\mu_x)(y-\mu_y)}{\eta}$$

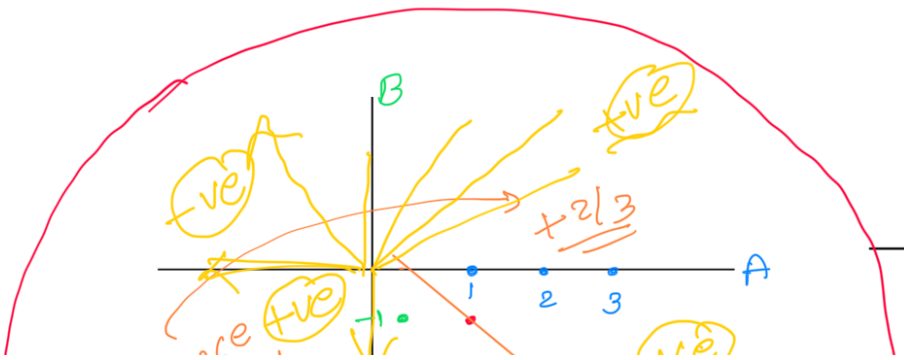
$-1 \quad -2 \quad -3 : x$

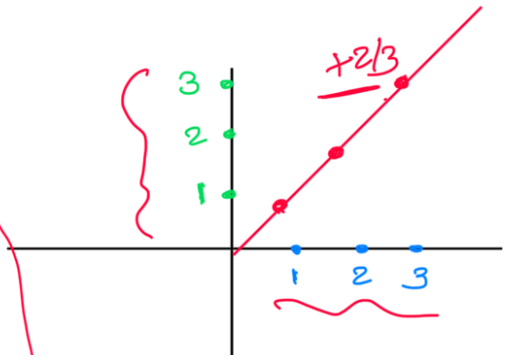
$-1 \quad -2 \quad -3 : y$

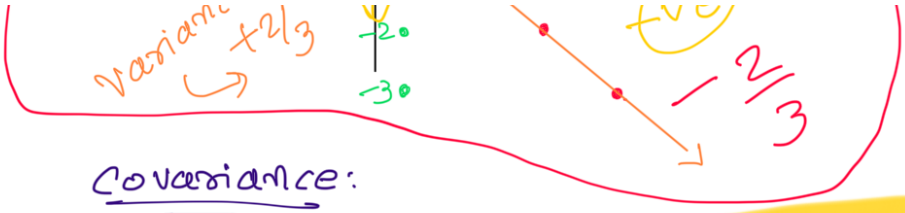
$\mu_x = -2$

$\mu_y = -2$

$$\frac{(-1-(-2))(-1-(-2))+(-2-(-2))^2+(-3-(-2))^2}{3} = 1+$$







Covariance:

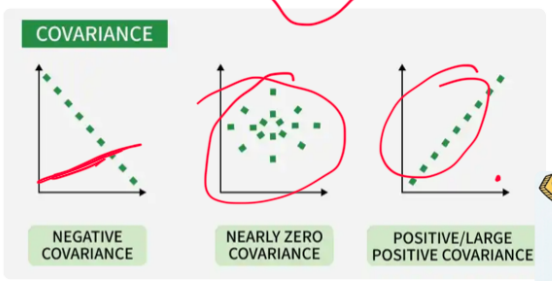
→ Describe the degree to which two variables are linearly Related

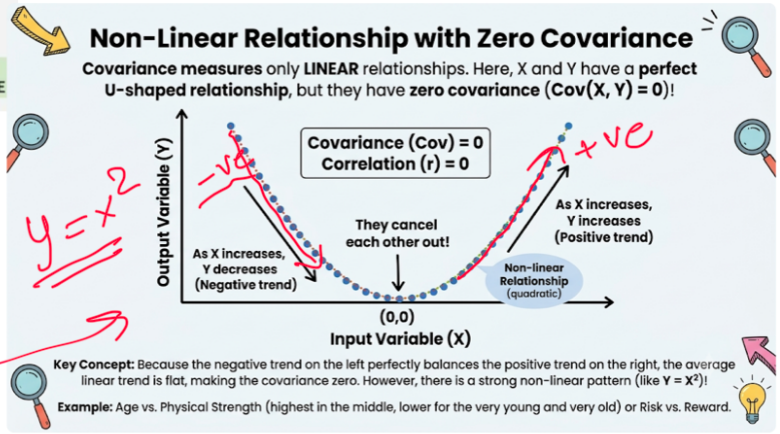
Covariance

+ve : move together in same direction

-ve : move in opposite direction

0 : Variables are not linearly Related. (there might be NON linear Relationship)





$y = x^2$

$Cov = 0$

→ Example where covariance = 0 but still there is non linear Relation ship.

$$Cov(x) = \frac{\Sigma (x - \mu_x)(x - \mu_x)}{}$$

$cov(x, x) = \frac{\sum (x - \mu_x)^2}{n} \Rightarrow$ Variance

$\pm(2/3) = \pm(2/3)$

→ limitation of covariance:

- only tells about the directional info like +ve, -ve
- only capture linear Relationship
- Does not tell about the strength of the Relation. only directional info.
- affectes by scale.

* corelation: (Direction + strength of Relationship)

$$\text{corelation} = \frac{cov(x,y)}{\sigma_x \sigma_y}$$

$-1 \leq cor \leq +1$

+1 = Perfect +ve Relation

0 = No corelation (No linear Relationship)

-1 = Perfect -ve Relation

corelation is std measure of the strength &

→ Correlation ...

Direction of the linear Relationship between two variables.

→ does not affect by scale.

Student	Height (m)	Height (cm)	Weight (kg)
A	1	100	50
B	2	200	60
C	3	300	70
Covariance with weight	10	1,000	—
Correlation with weight	1.0	1.0	—

Using population covariance.

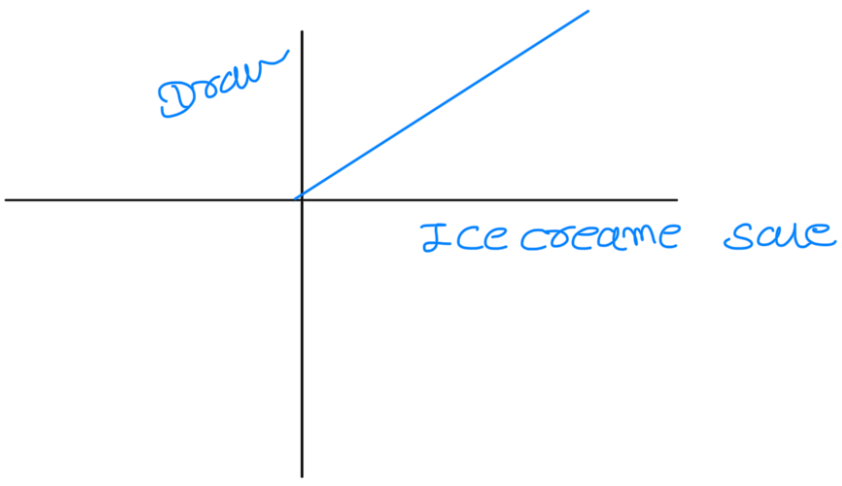
Teaching point: The relationship is exactly the same, but covariance changes from 10 to 1,000 just by changing meters to centimeters. Correlation stays at 1.

Therefore, **covariance magnitude alone cannot tell us the strength of a relationship**, while correlation provides a standardized measure between −1 and +1.

→ Correlation does not Imply Causation.

Ex: Icecream sale VS people die bcz of Drown

In summer



=) Use of coelation in ml:

